

Manual For Proposed Tool

This project implements a model-free evolutionary search tool for configuration performance tuning and compares it against a Random Search baseline using the datasets provided for the Intelligent Software Engineering coursework.

Repository Structure

The project contains the following files and folders:

- `datasets/` : dataset CSV files used for experimentation
- `search_results/` : generated CSV output files from `comparison.py` runs
- `visualization_results/` : generated boxplot images used in the report
- `tool.py` : implementation of the proposed evolutionary search tool
- `baseline.py` : implementation of the Random Search baseline
- `comparison.py` : runs both approaches for 30 repetitions and generates final comparison results
- `boxplot.py` : generates boxplots for result visualisation
- `final_results.csv` : summary of final experimental results
- `requirements.pdf` : required dependencies and library versions
- `manual.pdf` : instructions for running the tool
- `replication.pdf` : steps to reproduce the reported results

How to Run the Tool

Step 1: Navigate to the repository folder

Step 2: Run the proposed tool

```
python tool.py
```

This generates CSV files containing the search history for each system inside the `search_results` folder and prints the best configuration and final performance for each dataset.

Optional step 2.5: Run the baseline

```
Python baseline.py
```

This generates CSV files containing the search history for each system inside the `search_results` folder and prints the best configuration and final performance for each dataset

Step 3: Run `comparison.py` for comparing baseline and the proposed tool

```
python comparison.py
```

This runs both the proposed tool and the Random Search baseline for 30 repetitions using a fixed budget of 100 measurements and produces the result folders/files.

Step 4: Generate boxplots

```
python boxplot.py
```

This generates boxplot visualisations comparing the proposed tool and the baseline across repeated runs. The output images are stored in the `visualization_results` folder.

Outputs

The main output files are:

- `final_results.csv` : summary of average performance, standard deviation, improvement percentage, and p-values
- `*_raw_30_runs.csv` : performance values of tool and baseline across 30 repetitions.
- `*_boxplot.png` : boxplot figures used in the report

Notes

- Lower performance values indicate better configurations for all datasets
- Random seeds are fixed in `comparison.py` using `random.seed(42)` and `numpy.random.seed(42)` for reproducibility
- The measurement budget is fixed at 100 for fair comparison between the proposed tool and the baseline